

CSE 318: Artificial Intelligence Sessional

Online Task

A1

N-Puzzle with Obstacle Cell

Problem Statement

Modify your existing N-Puzzle solver by introducing one blocked cell in the board.

The blocked cell is represented by -1. No tile and no blank cell can ever move into this blocked cell.

For example, in a 4×4 board, the puzzle will contain:

- tiles from 1 to 14,
- one blank cell represented by 0,
- one blocked cell represented by -1.

So the total number of cells is still 16, but only 15 cells are usable.

Goal Checking

Update your goal checking so that the blocked cell is also taken into account.

For example, if the goal board is:

1	2	3	4
5	-1	6	8
9	10	7	12
13	14	11	0

Then a state is considered a goal only when all tiles, the blank, and the blocked cell are in their goal positions.

Heuristic Calculation

For Manhattan distance, compute the distance only for the numbered tiles:

$$h(n) = \sum_{\text{tiles } t} (|r_t - r_t^*| + |c_t - c_t^*|), \quad t \neq 0, t \neq -1$$